WELL-NIGH AFFINE IFF A FINITE FLAT GROUPOID PRESENTATION EXISTS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3338-3348 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 3350-3393, the complete author proof (42 body lines).

Editorial note. Navigation stubs: 9 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 24 of queue_stacks-stacks-more-morphisms.json (pb=2089, stmt 3338-3348, proof 3350-3393).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 04T4: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3393–3393. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04WY: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3393–3393. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06FH: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3393–3393. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06FI: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3393–3393. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02KB: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3393–3393. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04M9: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3393–3393. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DTY: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3393–3393. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04XD: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3393–3393. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04YC: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3393–3393. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. The following are equivalent*

(1) *$\mathcal{X}$ is well-nigh affine, and*
(2) *there exists a groupoid scheme (U, R, s, t, c) with U and R affine and $s, t : R \to U$ finite locally free such that $\mathcal{X} = [U/R]$.*

If true then $\mathcal{X}$ is quasi-compact, quasi-DM, and separated.

Proof. Assume $\mathcal{X}$ is well-nigh affine. Choose an affine scheme U and a surjective, flat, finite, and finitely presented morphism $U \to \mathcal{X}$. Set $R = U \times_{\mathcal{X}} U$. Then we obtain a groupoid (U, R, s, t, c) in algebraic spaces and an isomorphism $[U/R] \to \mathcal{X}$, see Algebraic Stacks, Lemma 04T4 and Remark 04WY. Since $s, t : R \to U$ are flat, finite, and finitely presented morphisms (as base changes of $U \to \mathcal{X}$) we see that s, t are finite locally free (Morphisms, Lemma 02KB). This implies that R is affine (as finite morphisms are affine) and hence (2) holds.

Suppose that we have a groupoid scheme (U, R, s, t, c) with U and R are affine and $s, t : R \to U$ finite locally free. Set $\mathcal{X} = [U/R]$. Then $\mathcal{X}$ is an algebraic stack by Criteria for Representability, Theorem 06FI (strictly speaking we don't need this here, but it can't be stressed enough that this is true). The morphism $U \to \mathcal{X}$ is surjective, flat, and locally of finite presentation by Criteria for Representability, Lemma 06FH. Thus we can check whether $U \to \mathcal{X}$ is finite by checking whether the projection $U \times_{\mathcal{X}} U \to U$ has this property, see Properties of Stacks, Lemma 04XD. Since $U \times_{\mathcal{X}} U = R$ by Groupoids in Spaces, Lemma 04M9 we see that this is true. Thus $\mathcal{X}$ is well-nigh affine.

Proof of the final statement. We see that $\mathcal{X}$ is quasi-compact by Properties of Stacks, Lemma 04YC. We see that $\mathcal{X} = [U/R]$ is quasi-DM and separated by Morphisms of Stacks, Lemma 0DTY. □